

Standard Operation Procedure

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This document details how to easily edit MARC records in bulk using OpenRefine and Excel and how to freely move and merge projects between these formats. (You will need MARCEdit for this).

An understanding of GREL (General Refine Expression Language) would be nice but is not necessary.

For this example, I am editing the 520 field (**but the process could be applied elsewhere**). The 520 field in a MARC record is a summary, abstract, annotation, or review of the material (book, audiobook, CD, ect..). It usually contains many spelling errors, making it a great example. I will use an example record set of 80 items.

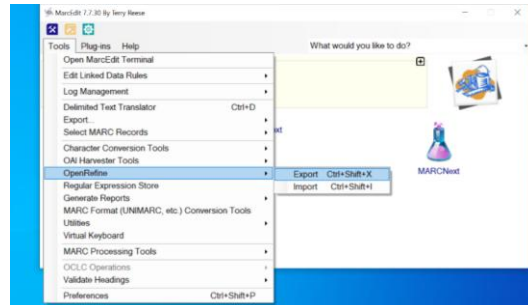
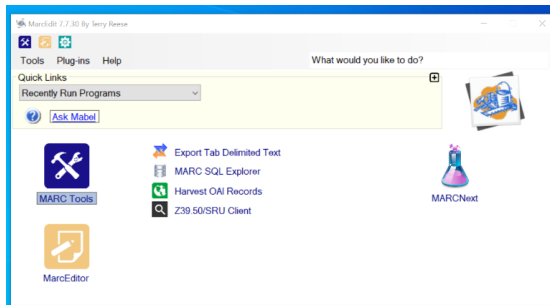
Quick overview: MarcEdit turns the mrc file into a format OpenRefine can support (json). Then, after importing the file into OpenRefine, I create a project and filter out the 520 fields (and other edits I may want to do). Then I export the file into an Excel to use the spell check feature to fix spelling errors. (Important note: OpenRefine also points out spelling errors, but only by clicking each individual cell. Excel jumps through cells in a column faster.). Once the Excel file has been edited, I re-import that excel into OpenRefine (though it makes a separate project). Then I merge into the original file. Now, with the fully edited original file, I re-import into MARC edit and create a mrc file.

In other words: MARCEdit-----> turn mrc into json---->OpenRefine project1-----> turn json into Excel-----> Excel-----> OpenRefine project2----> merge OpenRefine project1 and project2--->MARCEdit

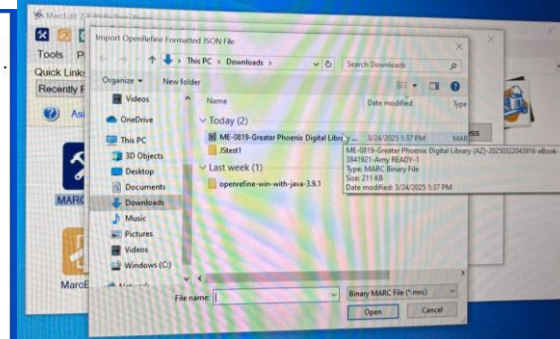
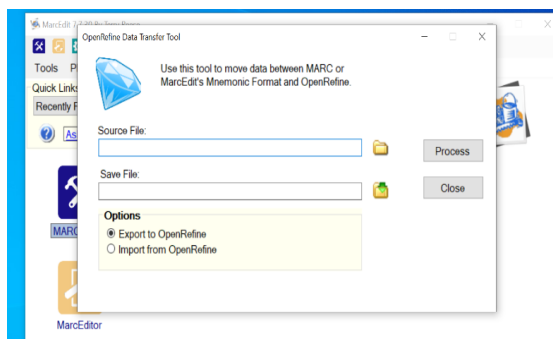
Step 1:

Open MarcEdit

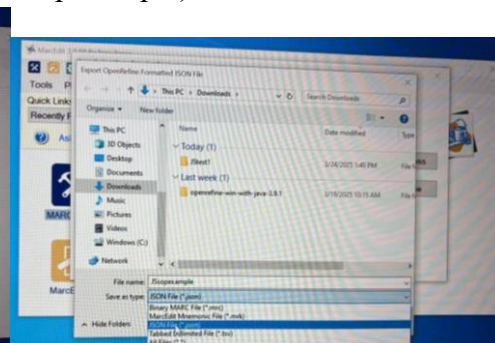
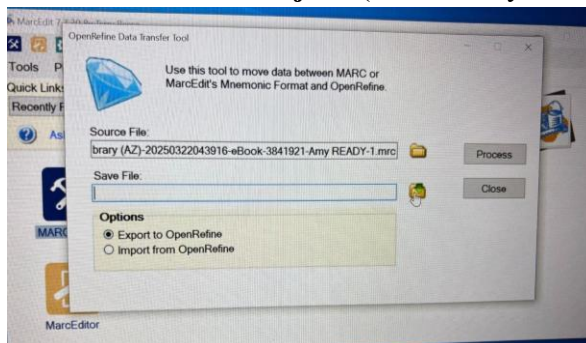
Click Tools, Click OpenRefine—>export



Go to Source File and select your record set



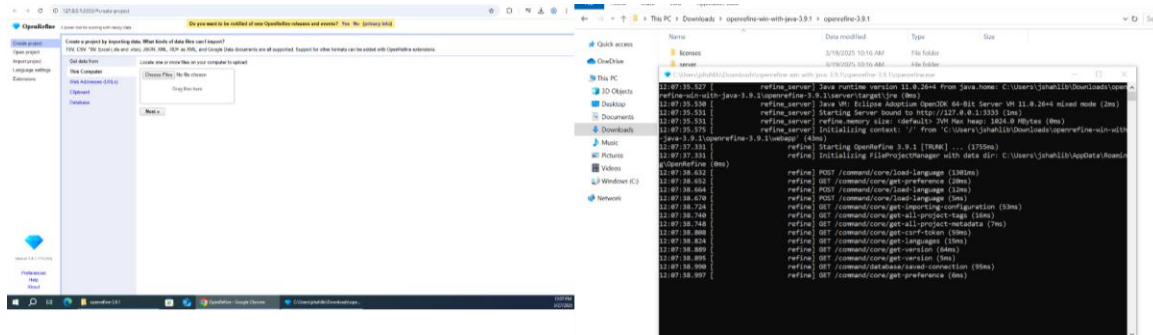
Name the file and save it as a **json** (I named my file JSsopexample)



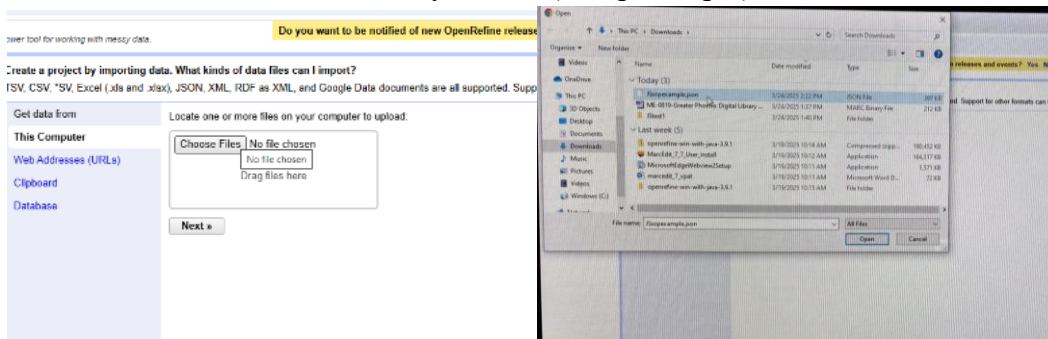
Click Process and the json file should show up in your downloads folder

Step 2:

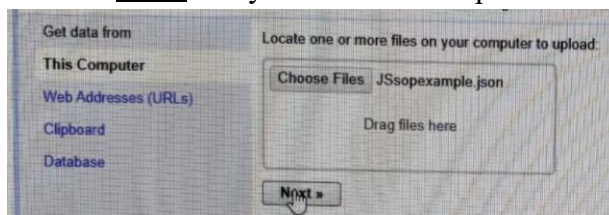
Open OpenRefine (a console should also open up, do not worry about it, it just records every action, do not close it but let it run in the background)



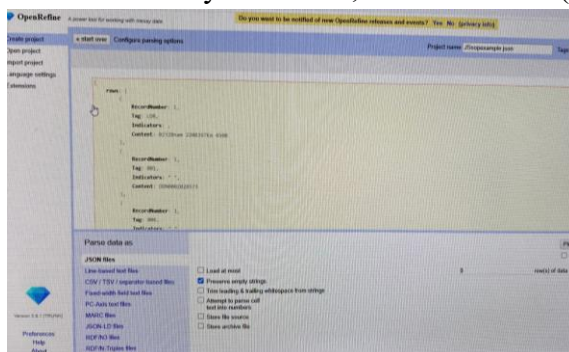
Press on Choose Files and select your file (JSsopexample)



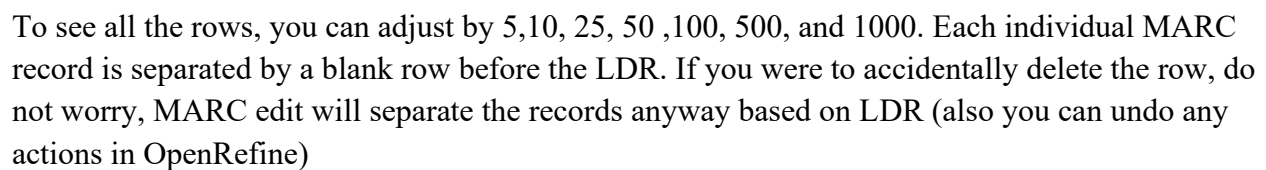
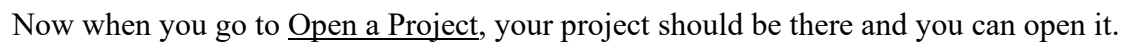
Click on Next and your file should upload



This should be your screen, select all rows (the yellow should encompass all):



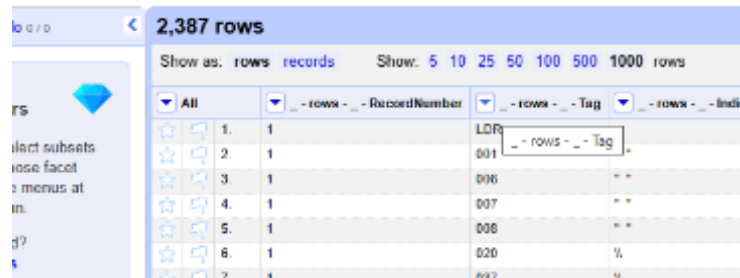
Click and this will be the following:



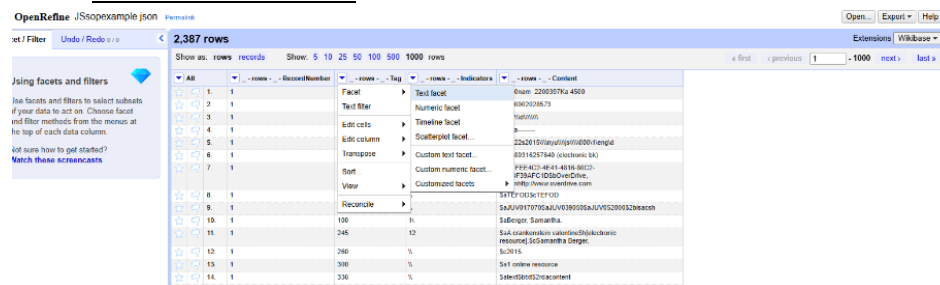
Step 3:

For this example, I will be filtering out the 520 fields but as mentioned before, you can filter out any fields. OpenRefine is known for their facets, and is very useful for looking at a group of any field at the same time.

Go to the row with the tags and click the dropdown arrow

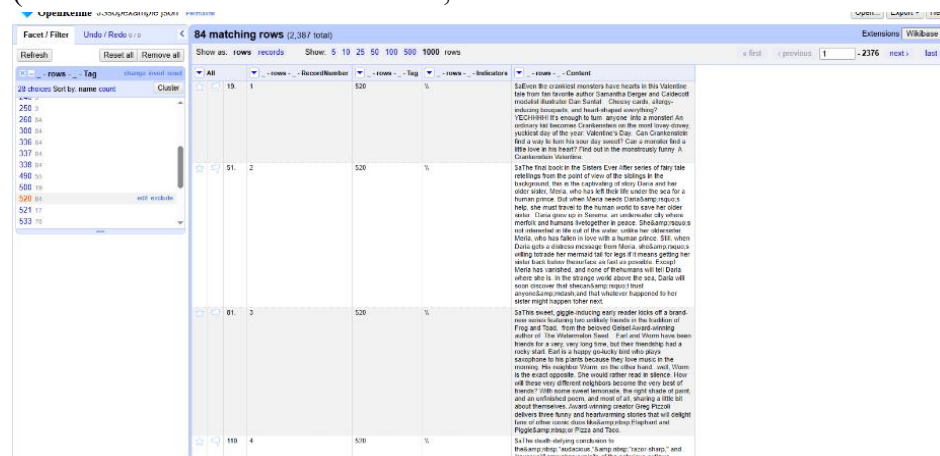


Go to Facet —>Text Facet



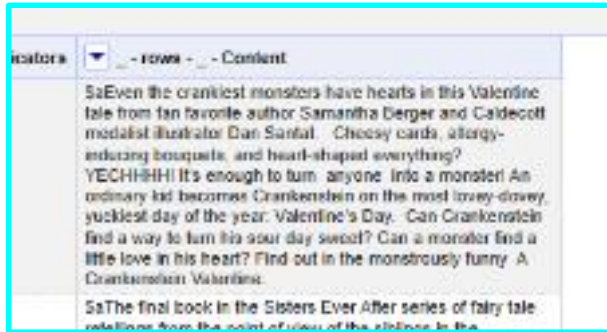
Select 520 (or whichever field you would like but for this example, 520)

As shown, only the 520 fields of all 80 records are filtered out to work with (the rest of the rows are still there, and can be shown if the filter is removed)

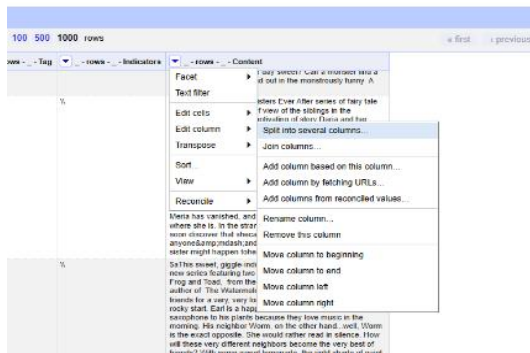


Now, we want to export this to Excel to spell check all of the content, but first we want to remove the \$a (a Blank Indicator Position) from the beginning of each content blurb and put it in

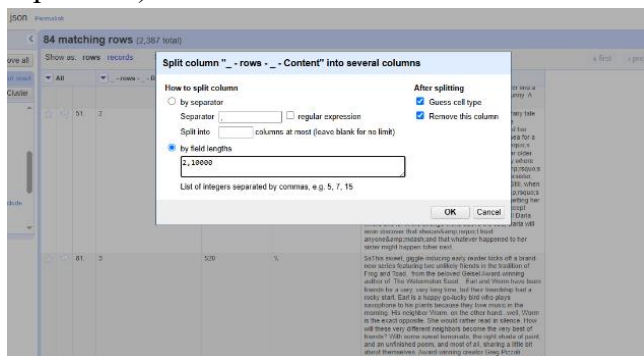
a separate column so spellcheck does not think it is a misspelling (do not worry, we will rejoin them later)



To remove the \$a (or any indicator) from the content and put in a separate column, press the dropdown arrow next to the “Content” column, select Edit Column—> Split into Several Columns



In the pop-up box, select By Field Length and enter 2,10000 (2 stands for the number of characters in \$a and 10000 stands for the rest of the characters. You can input 5000 or 6000 for whichever is the longest content blurb. It does not matter if you are unsure of the precise number of characters, overestimation is ideal. The 2 for first two characters is most important for the separation).



After pressing OK, you can see the separated columns (the column with the \$a is called “Content 1” and the column with the text is called “Content 2”).

	\$a	Content 1	Content 2
1	19	1	520
2	51	2	520
3	51	2	520
4	51	2	520
5	51	2	520
6	51	2	520
7	51	2	520
8	51	2	520
9	51	2	520
10	51	2	520
11	51	2	520
12	51	2	520
13	51	2	520
14	51	2	520
15	51	2	520
16	51	2	520
17	51	2	520
18	51	2	520
19	51	2	520
20	51	2	520
21	51	2	520
22	51	2	520
23	51	2	520
24	51	2	520
25	51	2	520
26	51	2	520
27	51	2	520
28	51	2	520
29	51	2	520
30	51	2	520
31	51	2	520
32	51	2	520
33	51	2	520
34	51	2	520
35	51	2	520
36	51	2	520
37	51	2	520
38	51	2	520
39	51	2	520
40	51	2	520
41	51	2	520
42	51	2	520
43	51	2	520
44	51	2	520
45	51	2	520
46	51	2	520
47	51	2	520
48	51	2	520
49	51	2	520
50	51	2	520
51	51	2	520
52	51	2	520
53	51	2	520
54	51	2	520
55	51	2	520
56	51	2	520
57	51	2	520
58	51	2	520
59	51	2	520
60	51	2	520
61	51	2	520
62	51	2	520
63	51	2	520
64	51	2	520

Go to the top right hand corner and press on Export. Select Excel and the Excel sheet should be downloaded.

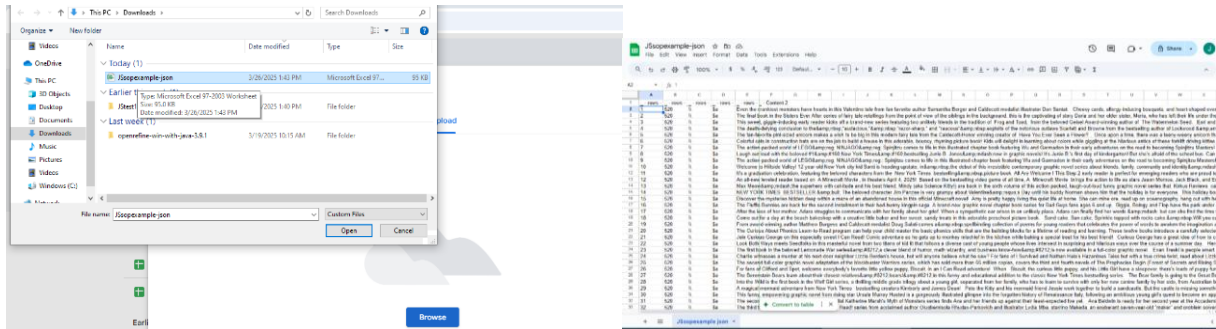
	\$a	Content 1	Content 2
1	19	1	520
2	51	2	520
3	51	2	520
4	51	2	520
5	51	2	520
6	51	2	520
7	51	2	520
8	51	2	520
9	51	2	520
10	51	2	520
11	51	2	520
12	51	2	520
13	51	2	520
14	51	2	520
15	51	2	520
16	51	2	520
17	51	2	520
18	51	2	520
19	51	2	520
20	51	2	520
21	51	2	520
22	51	2	520
23	51	2	520
24	51	2	520
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26	51	2	520
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28	51	2	520
29	51	2	520
30	51	2	520
31	51	2	520
32	51	2	520
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34	51	2	520
35	51	2	520
36	51	2	520
37	51	2	520
38	51	2	520
39	51	2	520
40	51	2	520
41	51	2	520
42	51	2	520
43	51	2	520
44	51	2	520
45	51	2	520
46	51	2	520
47	51	2	520
48	51	2	520
49	51	2	520
50	51	2	520
51	51	2	520
52	51	2	520
53	51	2	520
54	51	2	520
55	51	2	520
56	51	2	520
57	51	2	520
58	51	2	520
59	51	2	520
60	51	2	520
61	51	2	520
62	51	2	520
63	51	2	520
64	51	2	520

Now before we go to the Excel sheet, rejoin “Content 1” and “Content 2” by using the Undo / Redo function at the top left and selecting the very first step to revert to the original file. (This is crucial because when separating the columns with the facet, the other rows were still impacted and disarranged even when out of our view). **We need to revert back to the original.**

	\$a	Content 1	Content 2
1	19	1	520
2	51	2	520
3	51	2	520
4	51	2	520
5	51	2	520
6	51	2	520
7	51	2	520
8	51	2	520
9	51	2	520
10	51	2	520
11	51	2	520
12	51	2	520
13	51	2	520
14	51	2	520
15	51	2	520
16	51	2	520
17	51	2	520
18	51	2	520
19	51	2	520
20	51	2	520
21	51	2	520
22	51	2	520
23	51	2	520
24	51	2	520
25	51	2	520
26	51	2	520
27	51	2	520
28	51	2	520
29	51	2	520
30	51	2	520
31	51	2	520
32	51	2	520
33	51	2	520
34	51	2	520
35	51	2	520
36	51	2	520
37	51	2	520
38	51	2	520
39	51	2	520
40	51	2	520
41	51	2	520
42	51	2	520
43	51	2	520
44	51	2	520
45	51	2	520
46	51	2	520
47	51	2	520
48	51	2	520
49	51	2	520
50	51	2	520
51	51	2	520
52	51	2	520
53	51	2	520
54	51	2	520
55	51	2	520
56	51	2	520
57	51	2	520
58	51	2	520
59	51	2	520
60	51	2	520
61	51	2	520
62	51	2	520
63	51	2	520
64	51	2	520

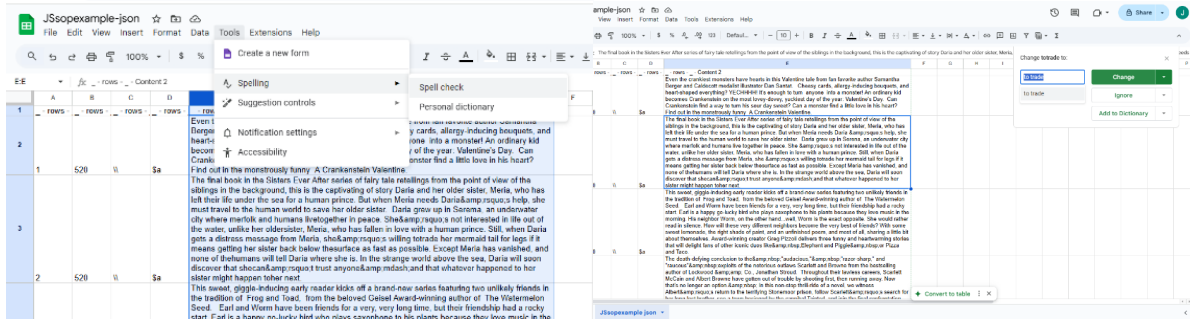
Step 4

Open your downloaded excel in Excel or Google Sheets (I am using Google Sheets)

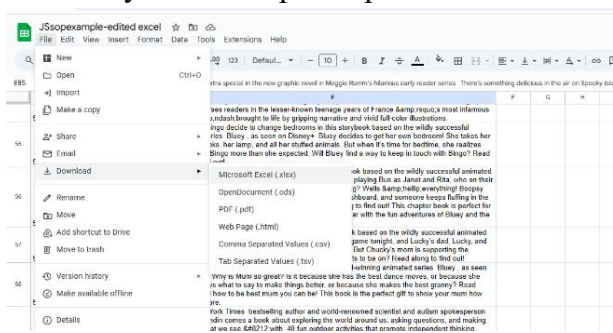


Re-adjust the formatting if wanted, but be wary of certain changes. Don't add extra rows or columns as we will need to re-merge this. I like to resize and text wrap the content column to view easier

Select the entire column and use the Spell Check tool which skips from cell to cell giving suggestions

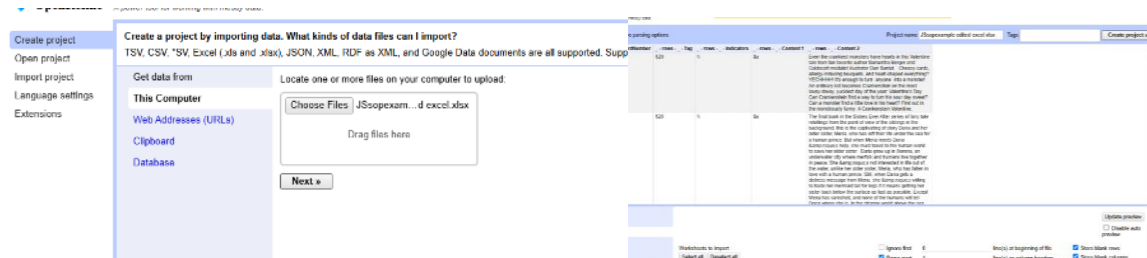


In Google Sheets, once you are finished with the desired edits, download the final product into a Microsoft Excel (changing the name so as not to confuse it with the original). For my example, I named my file “JSsopexample-edited excel”



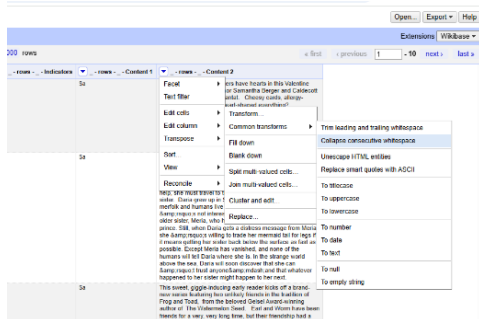
Step 5:

Using the same steps for uploading the record set into OpenRefine, upload the excel (JSsopexample-edited excel) and Create Project

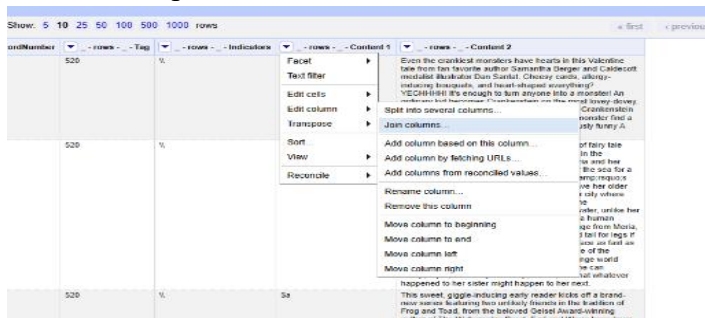


In this **new project**, there are some more adjustments to make before merging

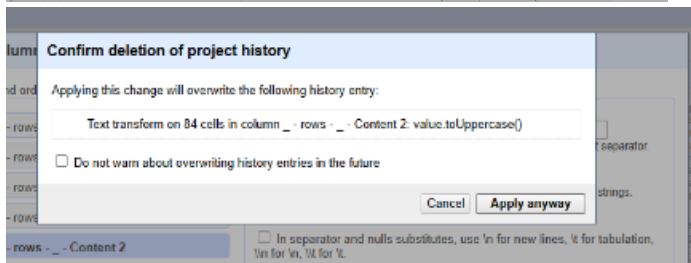
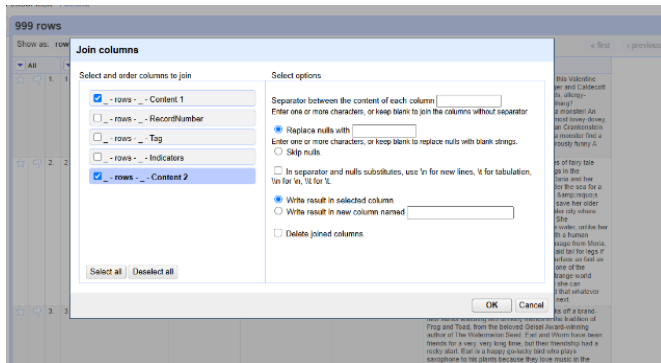
First, OpenRefine has some handy functions to clean up the content such as Collapsing Consecutive Whitespaces, ect... Go to the “Content 2” column (The column with the text, **not** the \$a), click on the dropdown arrow—>Edit Cells—> Common Transformations
There is a list of useful cleanup tools to use



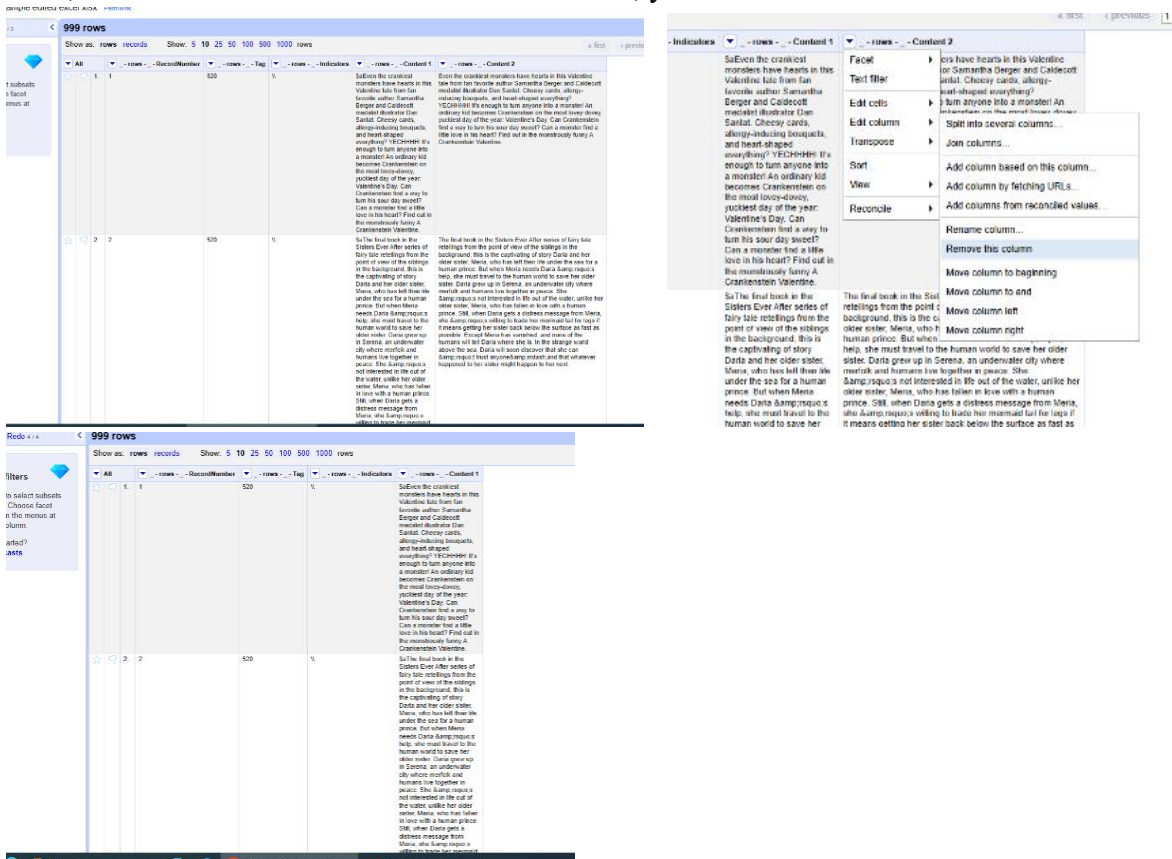
Second, you need to combine the “Content 1” and “Content 2” columns
Press the dropdown arrow on “Content 1” and select Edit Columns—>Join Columns



In the pop-up box, check the “Content 2” column box to join and select OK. You may get a histories warning, select Apply Anyway



Now, with the columns combined in “Content 1”, you can delete “Content 2” column



Step 6:

Now, for the explanation, I will be referring to the original record set as Project1 and the edited excel as Project2. We need to merge the **content** columns of Project1 and Project2.

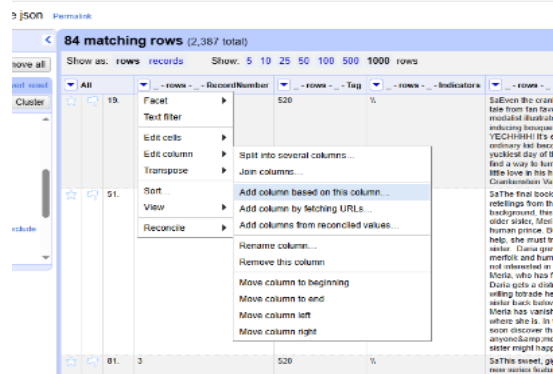
Project2 has the corrected content and Project1 is the original record set that has all of the other fields.

The first step is to identify a key. A key refers to a column that is the same in both projects (but not the same value throughout all the cells in the same project) and acts as a reference so OpenRefine knows what cells to match up in a column. In this instance, the key is the “Record Numbers” column which goes consecutively 1,2,3....



RecordNumber	Tag	Indicators
1	1	520
2	2	520

In the project into which data is being imported (**Project1**), select the column matching the key (“record numbers” column) and click on the dropdown arrow (make sure the 520 text facet filter is on). Select Edit Column—>Add a Column Based on this Column



In the pop-up window, give the new column a name and then enter this expression in the GREL expression box (**not the same exact expression, you will swap out the names in the quotes for names pertaining to your project**):

```
cell.cross('Project2','Key').cells['ImportColumn'].value[arg4]
```

- **Project2**= name of project you are exporting data from
- **Key**= name of the key column
- **ImportColumn**= name of the column you are importing
- **arg4** = indicate which value to import in the array (if multiple matches for the key) (recommended to use 0)

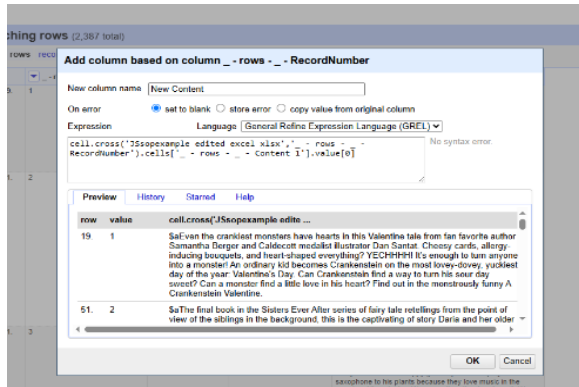
In my project for example, I named my new column “New Content” and inputted the following:

```
cell.cross('JSsopexample edited excel.xlsx','_ - rows - _ - RecordNumber').cells['_ - rows - _ - Content 1'].value[0]
```

- “**JSsopexample edited excel.xlsx**” is the name of Project2
- “**_ - rows - _ - RecordNumber**” is the column storing all of the record numbers
- “**_ - rows - _ - Content 1**” is the column storing all of the edited content (“Content 1” column in Project2)

Remember that regular expressions are CASE SENSITIVE. Nothing will happen if the column name and project name are not exact.

The “Preview” can help you double check which column you are importing



Press okay and the column will be imported next to the key column

84 matching rows (2,387 total)						Extensions W	
Show as: rows records Show: 5 10 25 50 100 500 1000 rows						first previous 1 -2376 next	
	All	_ - rows - _ - RecordNumber	New Content	_ - rows - _ - Tag	_ - rows - _ - Indicators	_ - rows - _ - Content	
19	1		SoEven the crankiest monsters have hearts in this Valentine tale from fan favorite author Samantha Berger and Caldecott medalist illustrator Dan Santat. Cheesy cards, allergy-inducing bouquets, and heart-shaped everything? YECHEHHH! It's enough to turn anyone into a monster! An ordinary kid becomes Crankenstein on the most lovey-dovey, yuckiest day of the year: Valentine's Day. Can Crankenstein find a way to turn his sour day sweet? Can a monster find a little love in his heart? Find out in the monstrously funny A Crankenstein Valentine.	520	%	SoEven the crankiest monsters have hearts in this Valentine tale from fan favorite author Samantha Berger and Caldecott medalist illustrator Dan Santat. Cheesy cards, allergy-inducing bouquets, and heart-shaped everything? YECHEHHH! It's enough to turn anyone into a monster! An ordinary kid becomes Crankenstein on the most lovey-dovey, yuckiest day of the year: Valentine's Day. Can Crankenstein find a way to turn his sour day sweet? Can a monster find a little love in his heart? Find out in the monstrously funny A Crankenstein Valentine.	
51	2		SoThe final book in the Sisters Ever After series of fairy tale retellings from the point of view of the siblings in the background, this is the captivating of story Daria and her older	520	%	SoThe final book in the Sisters Ever After series of fairy tale retellings from the point of view of the siblings in the background, this is the captivating of story Daria and her	

Step 7:

Now we are onto the final steps of merging the “New Content” with the original “Content” column in Project1 (you can close Project2 as it is no longer of use)

First, shift the “New Content” column to the right of the original “Content” column by pressing the dropdown arrow on the “New Content” and select Edit Column—> Move Column to Right

The screenshot shows a data table with columns: Facet, Text filter, Edit cells, Edit column, Transpose, Sort, View, Reconcile, and a 'New Content' column. A context menu is open over the 'New Content' column, listing actions such as 'Add column based on this column...', 'Add column by fetching URLs...', 'Add columns from reconciled values...', 'Rename column...', 'Remove this column', 'Move column to beginning', 'Move column to end', 'Move column left', and 'Move column right'. The 'Move column right' option is highlighted.

Then, on the original “Content” column, click the dropdown arrow and select Edit Column—> Join Columns

The screenshot shows the same data table, but now the 'New Content' column has been moved to the right of the 'Content' column. A context menu is open over the 'Content' column, listing actions such as 'Add column based on this column...', 'Add column by fetching URLs...', 'Add columns from reconciled values...', 'Rename column...', 'Remove this column', 'Move column to beginning', 'Move column to end', 'Move column left', and 'Move column right'. The 'Join Columns' option is highlighted.

Because we want the “New Content” text to completely replace the original “Content” text, in the pop-up box, check the “New Content” box and un-check the original “Content” box

The image shows two screenshots of a 'Join columns' dialog box in a software application. The dialog box is titled 'Join columns' and has a subtitle '84 matching rows (2,387 total)'. It contains two main sections: 'Select and order columns to join' and 'Select options'.

In the top screenshot, the 'Content' column is selected in the 'Select and order columns to join' section. In the 'Select options' section, the 'Replace nulls with' option is selected, and the 'Write result in selected column' option is selected.

In the bottom screenshot, the 'New Content' column is selected in the 'Select and order columns to join' section. In the 'Select options' section, the 'Replace nulls with' option is selected, and the 'Write result in selected column' option is selected.

Press OK

Now, the original “Content” column should have the newly edited and spell checked text!

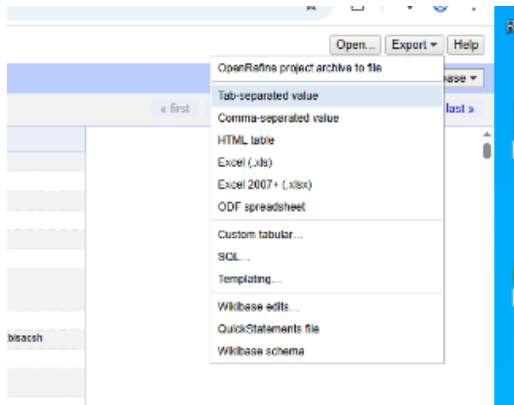
Delete the “New Content” column as there is no more use for it

Now you have a fully completed record set!!

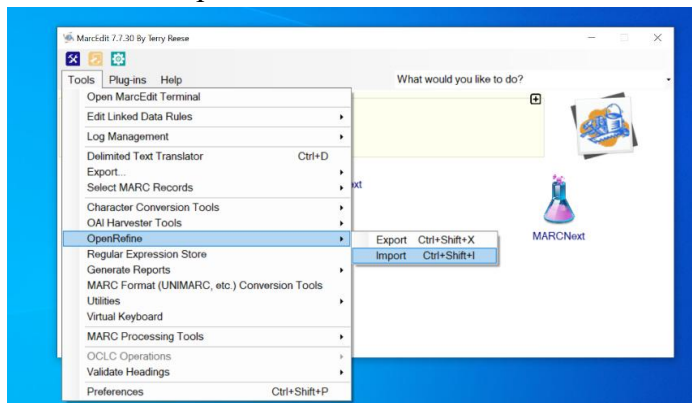
Step 8:

Now, with the fully edited original file, we need to re-import into MARC edit and create a mrc file.

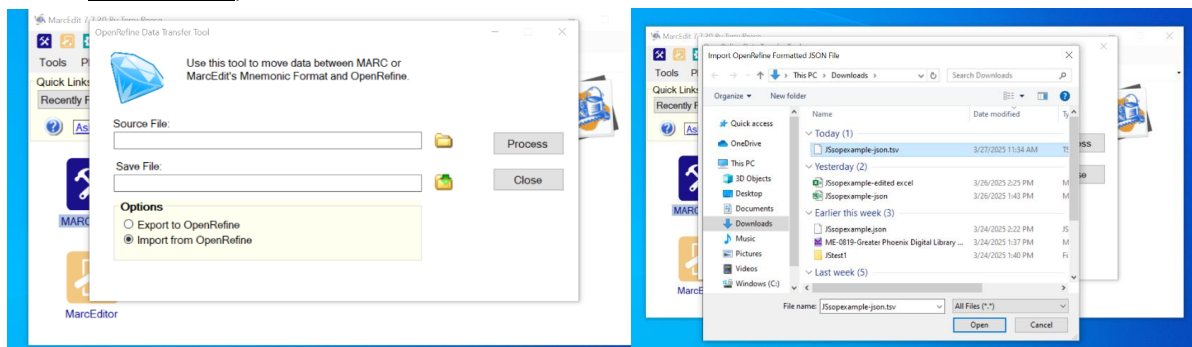
In Project1, at the top right of OpenRefine, Select Export—>Tab-Separated value
The tsv file should download



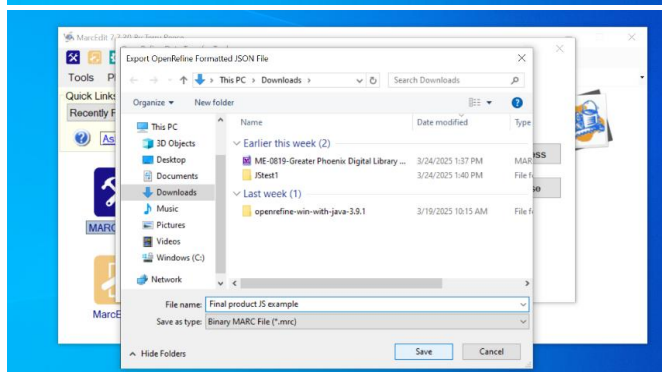
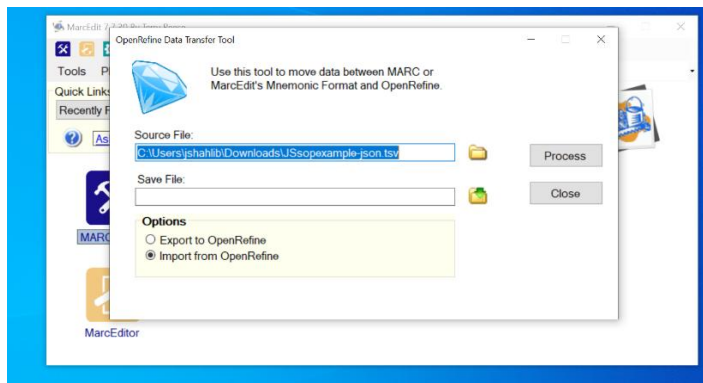
Like on step 1, open MarcEdit, select Tools---> OpenRefine
Then select Import



In the Source File, select the tsv



Now you can save the file as a mrc (or mrk if you want to do more work in MarcEdit) and click Process



Your final product is ready!

Be sure to delete all unnecessary files created in this process

Citations/ Supplementary Resources

tpreese. "Importing and Exporting Data between MarcEdit and OpenRefine." *YouTube*, 16 Jan. 2016, youtu.be/ItZShMoZmWg?si=GPXqMPkeDODuAxJJ.

"Libguides: OpenRefine: Joining Projects." *Joining Projects - OpenRefine - LibGuides at University of Illinois at Urbana-Champaign*, 8 Nov. 2023, guides.library.illinois.edu/openrefine/joiningprojects.

"Reconciling: OpenRefine." *OpenRefine RSS*, 5 June 2024, openrefine.org/docs/manual/reconciling.

Margaret, Post author By, et al. "A Librarian's Guide to OpenRefine." *Glorious Generalist*, www.gloriousgeneralist.com/2013/05/a-librarians-guide-to-openrefine/. Accessed 27 Mar. 2025.